

Your Goal

Your job is to design the **Endpoint Security** for a telecom company of 550 people (500 employees, 50 contractors) who use 30 apps in a hybrid (on-premise + cloud) environment.

"Endpoint" just means any device that connects to your company's network. This includes:

- Laptops (company-owned)
- Desktops
- Mobile Phones (iPhones, Androids)
- Contractor Laptops (a *huge* risk!)
- Servers (both on-premise and in the cloud)

Here's how to build your PDF, step-by-step.

Step 1: Define Your Scope (The "What")

First, state exactly *what* you are protecting. This shows you've read the company profile.

In your PDF, create a section called "Endpoint Scope" and make a table:

Device Type	User Group	Ownership	Estimated Count
Laptops (Windows/Mac)	Employees	Company-Owned	450
Desktops	Employees	Company-Owned	50
Mobile Phones	Employees	BYOD (Bring Your Own Device)	500
Laptops	Contractors	BYOD (Contractor-Owned)	50
Servers (On-Premise)	Infrastructure	Company-Owned	(Estimate 30)
Servers (Cloud)	Infrastructure	Company-Owned	(Estimate 20)

Why this is important: This list immediately highlights the two biggest problems:

1. **BYOD Mobiles:** You need to protect company data on *personal* phones.
2. **Contractor Laptops:** You have *no control* over these, so you must *force* them to be secure before they can connect.

Step 2: Identify Key Risks (The "Why")

Next, list the top 3-5 threats you're trying to prevent. This shows you understand *why* this matters.

In your PDF, create a section called "Key Security Risks":

- **Risk 1: Malware/Ransomware:** A contractor plugs in their infected personal laptop and

encrypts a critical company server.

- **Risk 2: Data Theft:** An employee on their last day (using their personal phone) copies the entire customer contact list from the cloud and emails it to themselves.
 - **Risk 3: Unpatched Devices:** A remote employee's laptop is 6 months out-of-date. It gets a virus, which then spreads to other systems when they connect.
 - **Risk 4: Lost/Stolen Device:** A laptop with unencrypted company data is stolen, leading to a massive data breach and regulatory fines (especially in Telecom!).
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Step 3: Design Your Security Solution (The "How")

This is the main part of your project. Break your design into three layers. This is how you will **implement Zero Trust**.

Zero Trust Principle: "Never Trust, Always Verify." It means you don't trust any device (even an employee's) just because it's "inside" the network. You must verify *every* endpoint.

In your PDF, create a section called "Endpoint Security Design":

3.1. Layer 1: The Core Tools (Protection)

These tools go on *every* device.

- **Endpoint Detection & Response (EDR):** This is "Next-Gen Antivirus." It doesn't just block known viruses; it watches for *suspicious behavior* (like a Word doc trying to encrypt files) and stops it.
- **Full-Disk Encryption (FDE):** Mandatory for ALL laptops (employee and contractor). If a laptop is stolen, the data is unreadable. Use BitLocker (Windows) and FileVault (Mac).
- **Host-Based Firewall:** A software firewall on every device to block unwanted network connections.

3.2. Layer 2: The Management (Control)

These tools give you *control* over all 550+ devices.

- **Unified Endpoint Management (UEM) / Mobile Device Management (MDM):** This is your **most important tool**. It's a single dashboard to manage *all* endpoints: laptops, phones, and contractor devices.
 - **For Laptops:** It pushes security policies, installs the EDR, and forces software updates.
 - **For Mobile (BYOD):** It uses "**Containerization**." It creates a secure "work" folder on the personal phone. All company apps (email, cloud apps) *live inside this container*. You can wipe the container without touching their personal photos.
- **Patch Management:** Your UEM tool will *force* all devices to install security patches within 7 days. No more unpatched systems!

3.3. Layer 3: The Policies (Rules)

This is how you handle the specific user groups.

- **Policy for Employees:**
 1. All company laptops are encrypted and have EDR.
 2. To access company data on a personal phone, the UEM "container" app *must* be installed.
 - **Policy for Contractors (This is your "A+" Answer):**
 1. Contractors *cannot* access any company data on their personal laptops.
 2. Instead, they must use a **Virtual Desktop Infrastructure (VDI)**.
 3. **How VDI works:** The contractor uses their laptop to log into a "virtual" Windows desktop that is running in your company's cloud. All 30 applications and all the data *stay inside the cloud*. The contractor's laptop is just a "dumb screen." They cannot copy data, and their infected laptop can't touch your network. This is perfect Zero Trust.
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Step 4: Address the "Exercise Considerations" (The "Extra Marks")

Show the professor you read the whole instruction sheet. Make a final section in your PDF.

- **Budget:** Our design saves money by focusing on a **cloud-based UEM**, which is cheaper than buying on-premise hardware. We prioritize the VDI for contractors, as they are the highest risk.
- **Integration (Hybrid):** A cloud-based UEM and EDR are *perfect* for a hybrid model. They work the same whether the employee is in the office (on-premise) or at home (cloud).
- **Scalability:** VDI and UEM are cloud services. They can easily scale from 550 users to 5000 with no new hardware.
- **User Experience vs. Security:** We balance this. The "container" on BYOD phones is less intrusive, and VDI is easy for contractors to use.
- **Contractor Access:** Solved perfectly with the VDI solution.
- **Zero Trust:** Our *entire model* is Zero Trust. We verify the *device* (with UEM/EDR) and isolate the *session* (with VDI) before granting any access.
- **Regulatory Compliance (Telecom):** Telecoms have strict data privacy rules. Our FDE (Encryption) and VDI (no data on contractor laptops) are essential for passing a compliance audit.

Your Final PDF Structure:

1. **Cover Page** (Group Name, Topic: Endpoint Security)

2. **Executive Summary** (1 paragraph: "We designed a Zero Trust endpoint solution...")
3. **Endpoint Scope** (Your table from Step 1)
4. **Key Security Risks** (Your list from Step 2)
5. **Proposed Security Design** (The 3 layers from Step 3)
 - 3.1. Core Tools (EDR, Encryption)
 - 3.2. Management (UEM/MDM)
 - 3.3. Key Policies (Employee, Contractor VDI)
6. **Addressing Project Requirements** (Your list from Step 4)
7. **Conclusion**